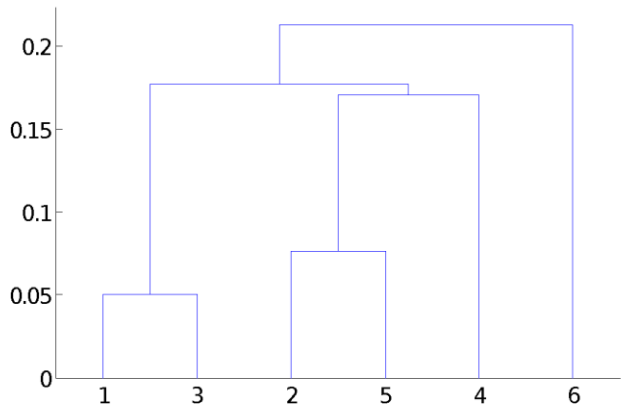
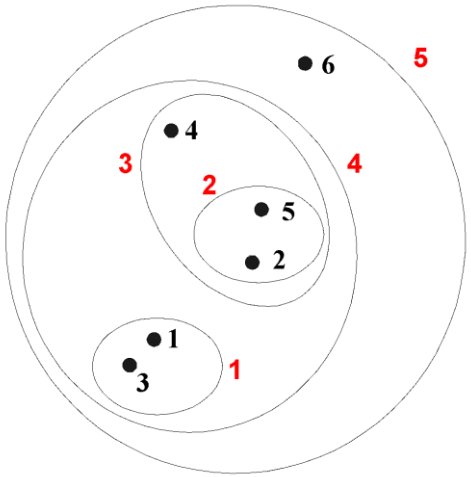
Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
  - A tree like diagram that records the sequences of merges or splits





Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by ‘cutting’ the dendogram at the proper level

- It provides detailed information about which observations are most similar to each other.
  - This level of detail is not provided by many other algorithms, which generally just return the ID of the cluster a given observation belongs to.

Hierarchical Clustering

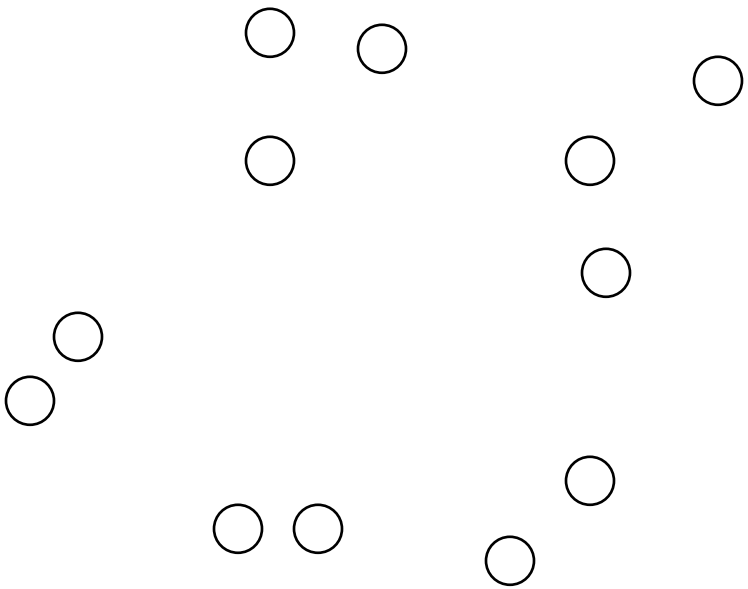
- Two main types of hierarchical clustering
  - Agglomerative: (bottom up)
    - Start with the points as individual clusters
    - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
    - It's also known as AGNES (Agglomerative Nesting)
  - Divisive: (top down)
    - Start with one, all-inclusive cluster
    - At each step, split a cluster until each cluster contains a point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
  - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
  1. Compute the proximity matrix
  2. Let each data point be a cluster
  3. **Repeat**
  4. Merge the two closest clusters
  5. Update the proximity matrix
  6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

Starting Situation

- Start with clusters of individual points and a proximity matrix

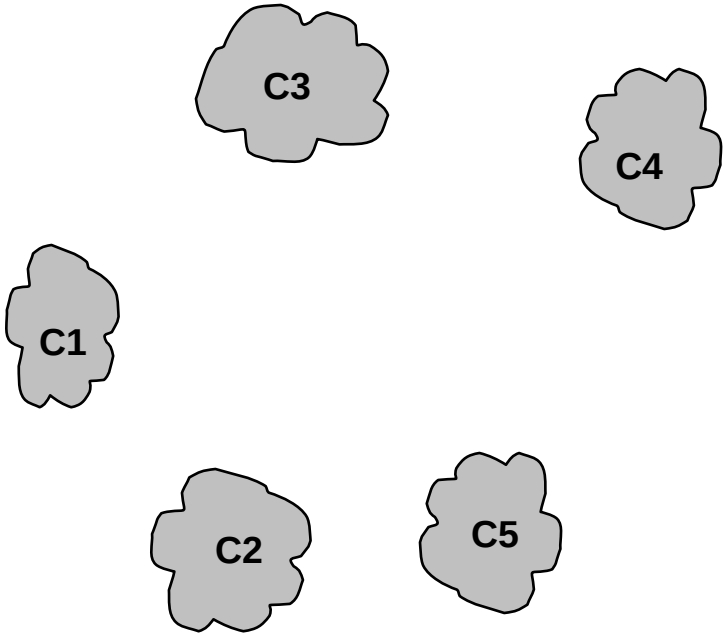


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

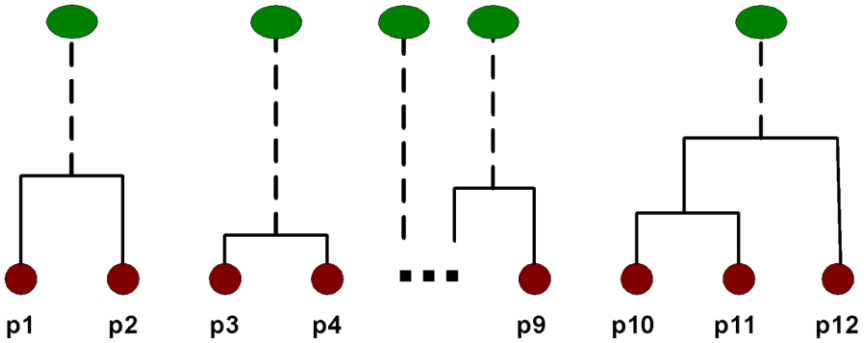
Intermediate Situation

- After some merging steps, we have some clusters



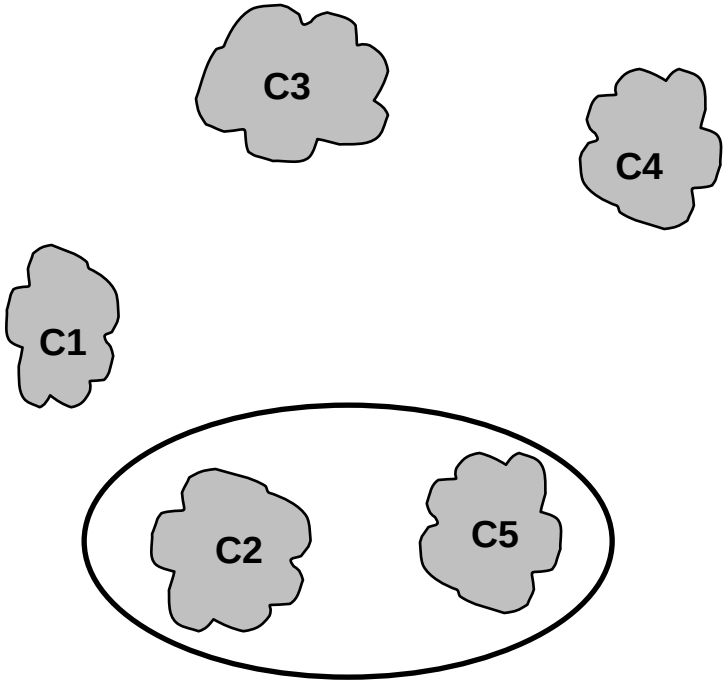
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



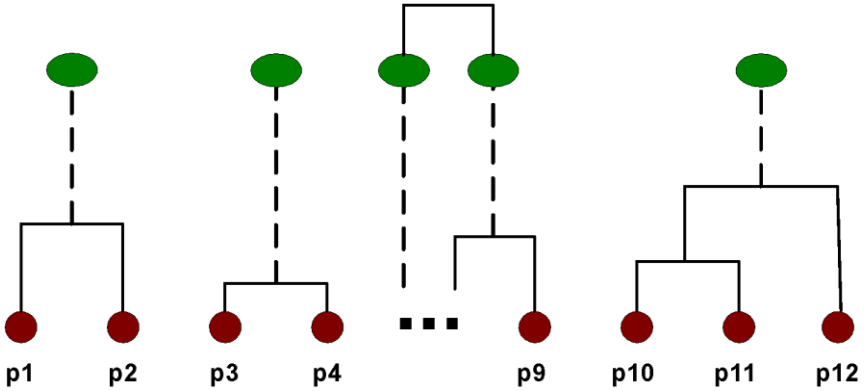
Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



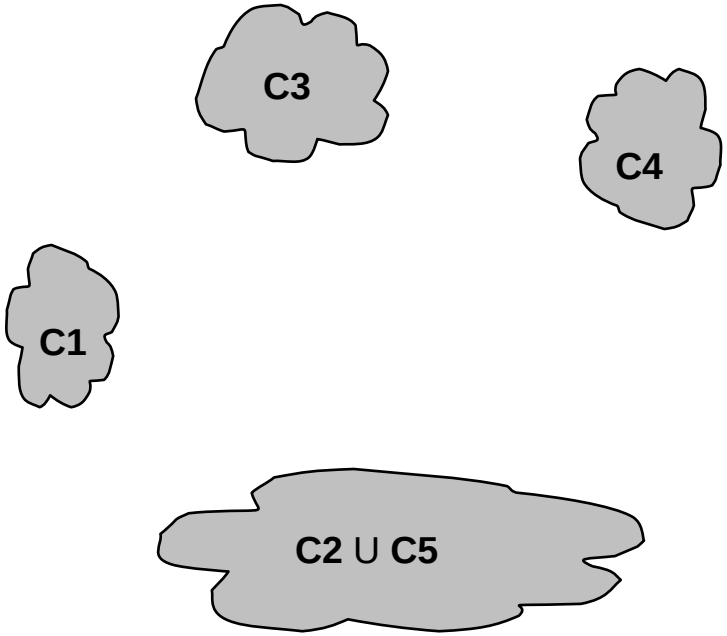
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



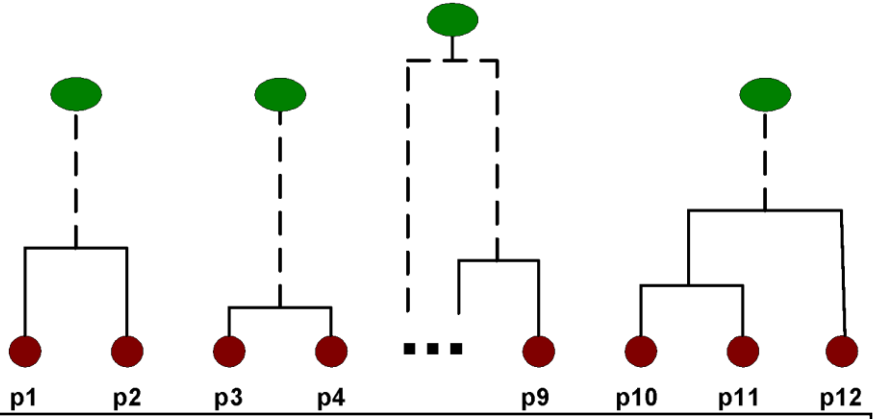
After Merging

- The question is “How do we update the proximity matrix?”

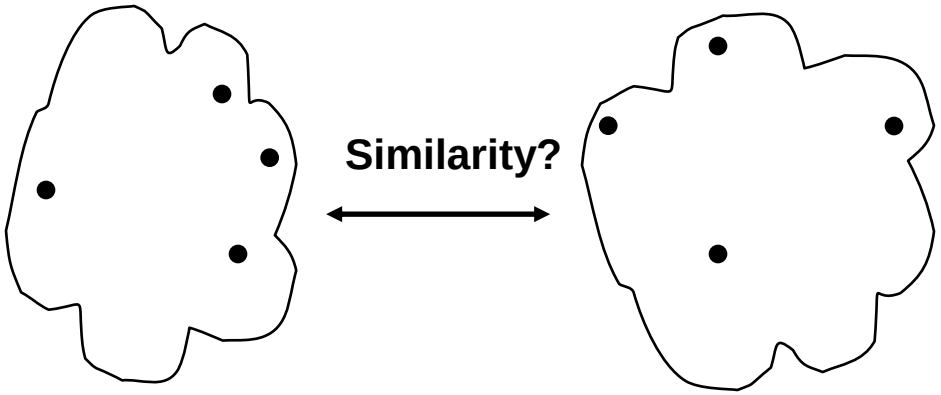


	C1	C2 ∪ C5	C3	C4
C1		?		
C2 ∪ C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



How to Define Inter-Cluster Similarity

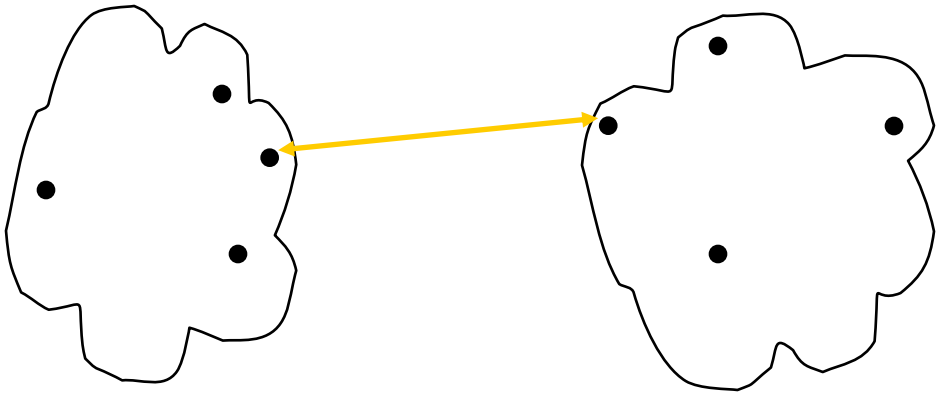


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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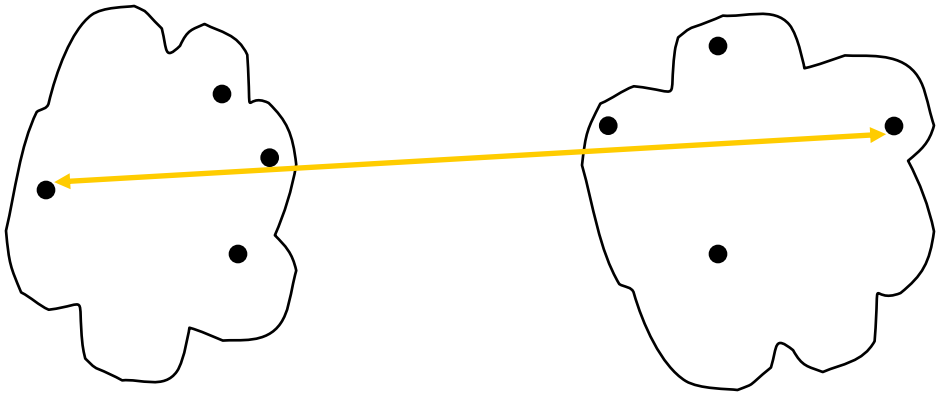
How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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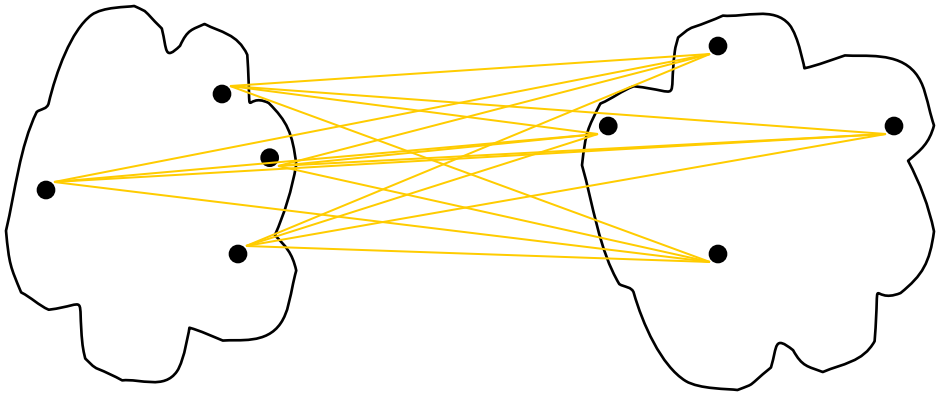
How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
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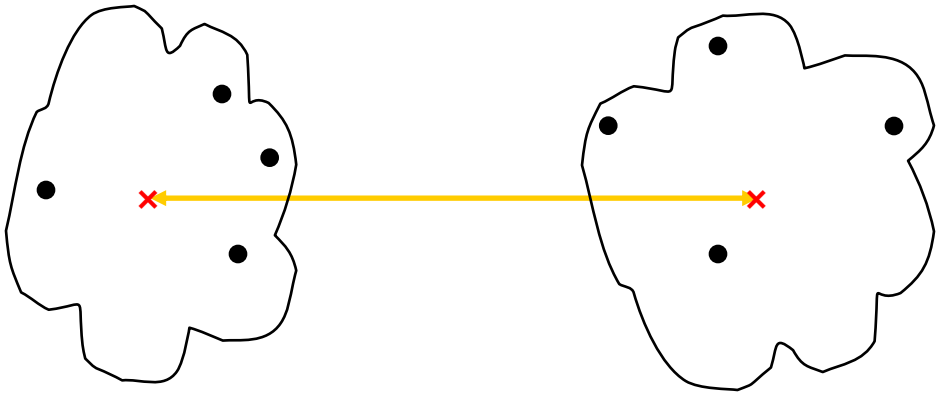
How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

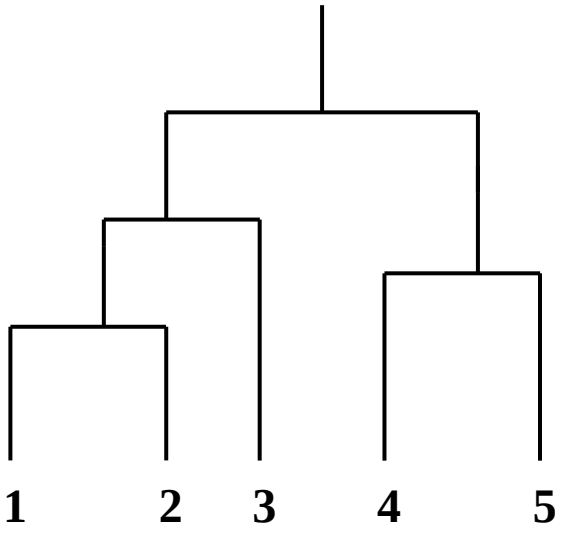
- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

Cluster Similarity: MIN or Single Link

https://www.youtube.com/watch?v=RdT7bhm1M3E

- Similarity of two clusters is based on the two most similar (closest) points in the different clusters
  - Determined by one pair of points, i.e., by one link in the proximity graph.

	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Cluster Similarity: MIN or Single Link

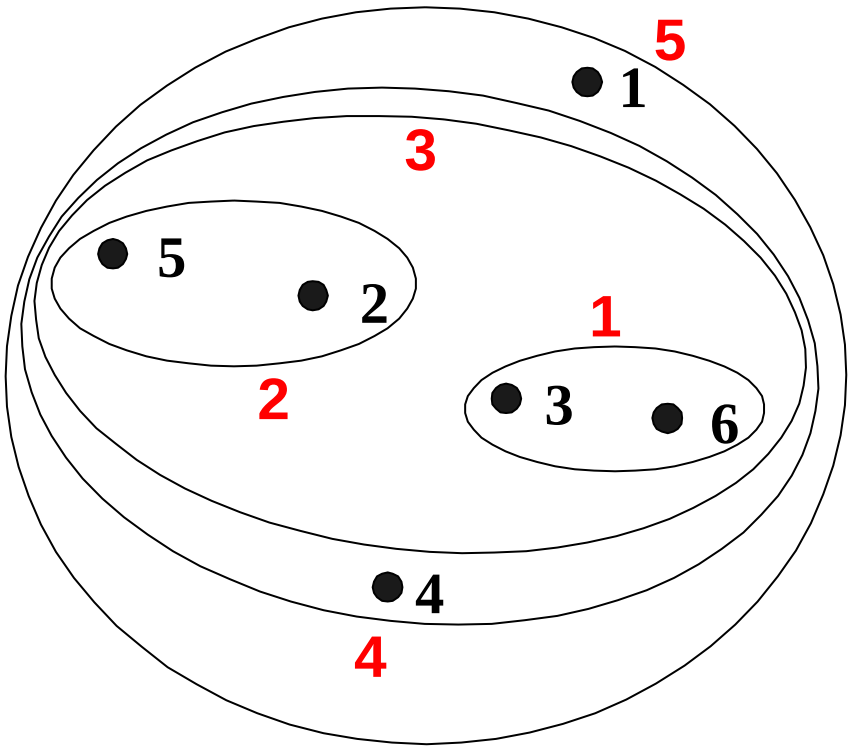
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

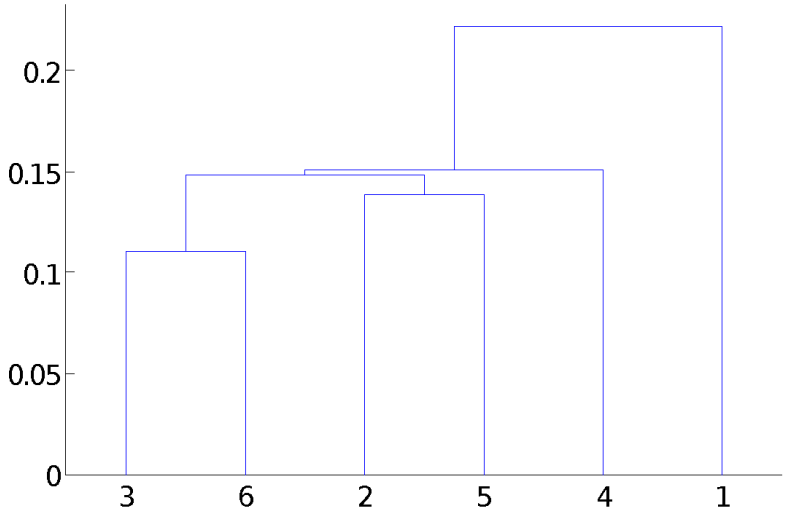
Euclidean distance matrix for 6 points.

Hierarchical Clustering: MIN

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.





Nested Clusters

Dendrogram

Cluster Similarity: MIN or Single Link

Recalculating matrix for cluster p3,p6

Dist (p3,p6) (p1)=min {(p3,p1) (p6, p1)}
Dist (p3,p6) (p2)=min {(p3,p2) (p6, p2)}
Dist (p3,p6) (p4)=min {(p3,p4) (p6, p4)}

$$
\begin{aligned}
dist(\{3,6\},\{2,5\}) &= \min(dist(3,2), dist(6,2), dist(3,5), dist(6,5)) \\
&= \min(0.15, 0.25, 0.28, 0.39) \\
&= 0.15.
\end{aligned}
$$

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.

- To update the distance matrix MIN[dist(P3,P6),P1)]
- MIN(dist(P3,P1), (P6,P1))

  = min[(0.22,0.23)]

  = 0.22
- To update the distance matrix MIN[dist(P3,P6),P2)]
- MIN(dist(P3,P2), (P6,P2))

  = min[(0.15,0.25)]

  = 0.15
- To update the distance matrix MIN[dist(P3,P6),P4)]
- MIN(dist(P3,P4), (P6,P4))

  = min[(0.15,0.22)]

  = 0.15
- To update the distance matrix MIN[dist(P3,P6),P5)]
- MIN(dist(P3,P5), (P6,P5))

  = min[(0.28,0.39)]

  = 0.28

The updated distance matrix for cluster P3, P6

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

Again find the min value in the updated matrix: P2, p5

- To update the distance matrix MIN[dist(P2,P5),P1)]
- MIN[dist(P2,P1), (P5,P1)]

  = min[(0.23,0.34)]

  = 0.23
- To update the distance matrix MIN[dist(P2,P5),(P3,P6)]
- MIN[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = min[(0.15,0.28)]

  = 0.15

To update the distance matrix MIN[dist(P2,P5),P4)]

MIN[dist(P2,P4), (P5,P4)]

= min[(0.20,0.29)]

= 0.20

The updated distance matrix for cluster P2,P5

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

Now again find the min value in the matrix: its found between p2,p5 and p3,p6

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P1]
- MIN[dist((P2,P5),P1), ((P3,P6),P1)]

  = min[(0.23,0.22)]

  = 0.22

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P4]
- MIN[dist((P2,P5),P4), ((P3,P6),P4)]

  = min[(0.20,0.15)]

  = 0.15

- The updated distance matrix for cluster P2,P5,P3,P6

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

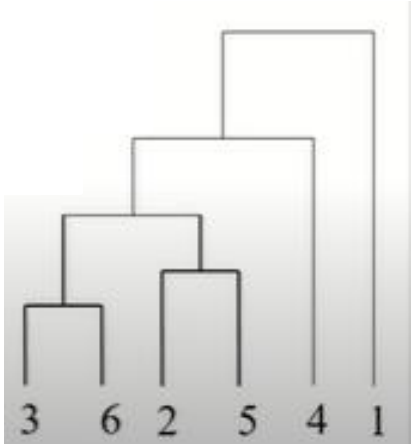
	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

- To update the distance matrix MIN[dist(P2,P5,P3,P6),P4]
- MIN[dist((P2,P5,P3,P6),P1), (P4,P1)]

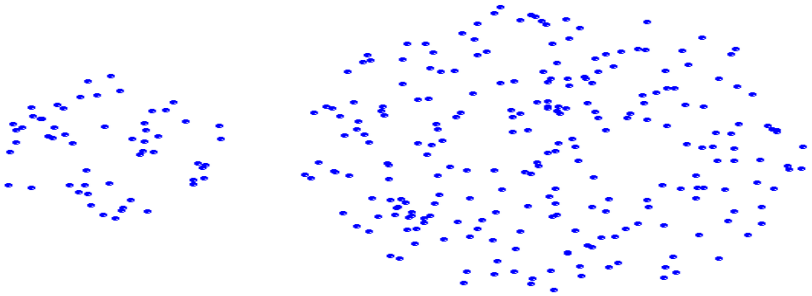
  = min[(0.22,0.37)]

  = 0.22

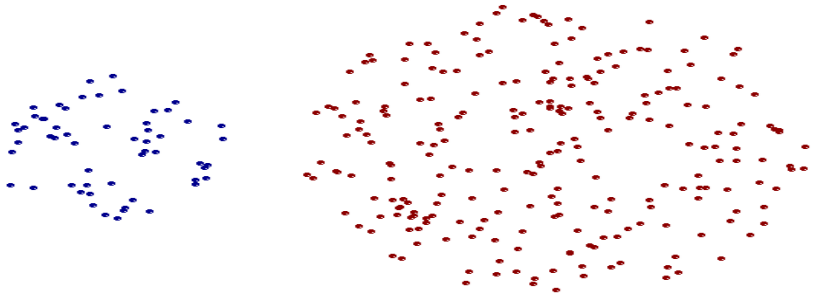
	P1	P2,P5,P3,P6,P4
P1	0	
P2,P5,P3,P6,P4	0.22	0



Strength of MIN



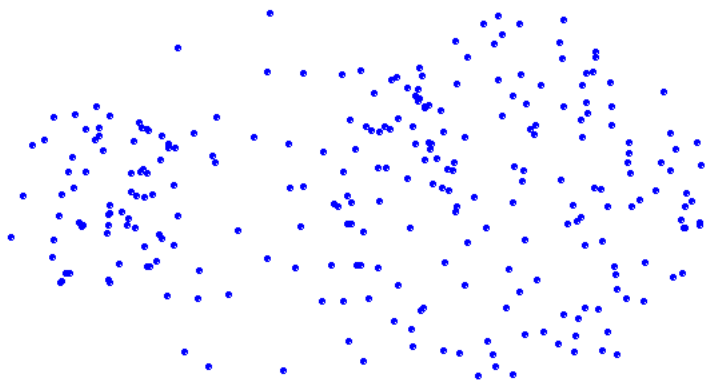
Original Points



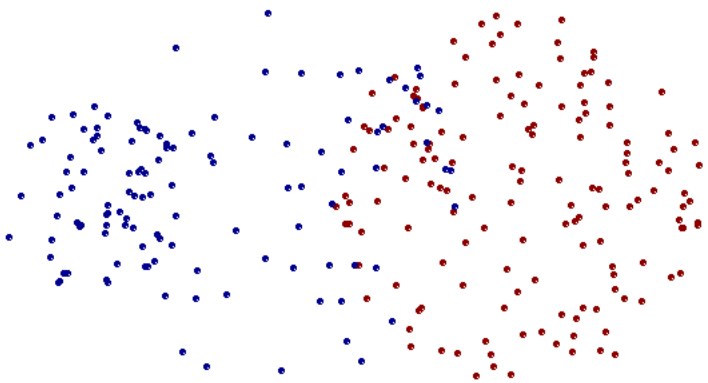
Two Clusters

- Can handle non-elliptical shapes

Limitations of MIN



Original Points



Two Clusters

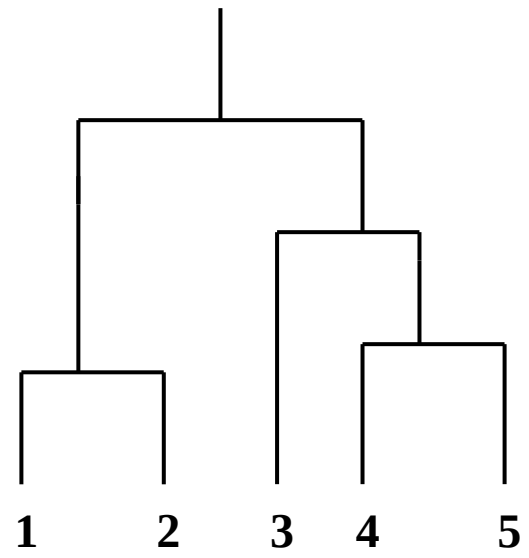
- **Sensitive to noise and outliers**

Cluster Similarity: MAX or Complete Linkage

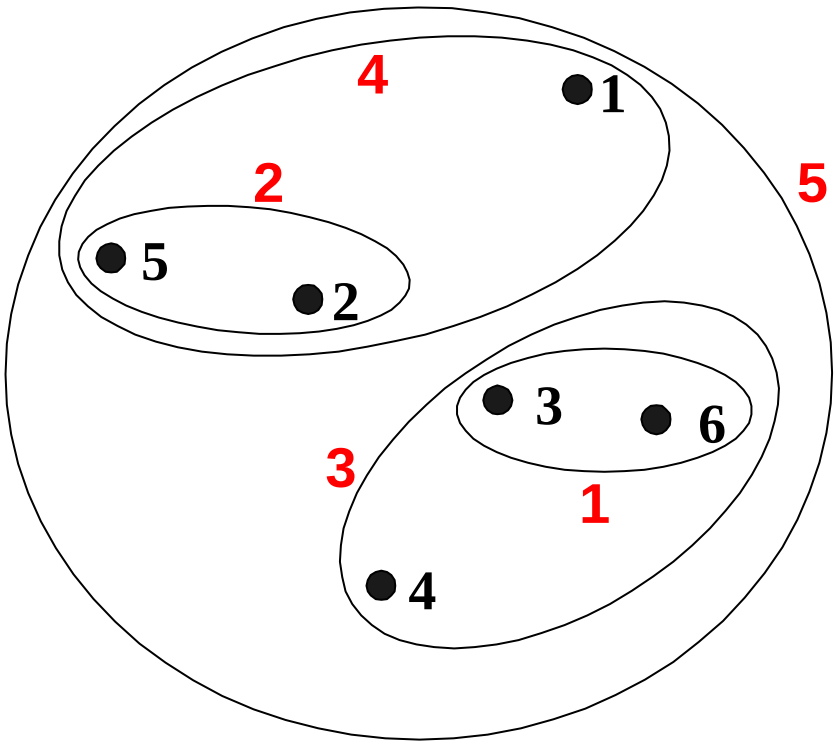
https://www.youtube.com/watch?v=Cy3ci0Vqs3Y

- Similarity of two clusters is based on the two least similar (most distant) points in the different clusters
  - Determined by all pairs of points in the two clusters

	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



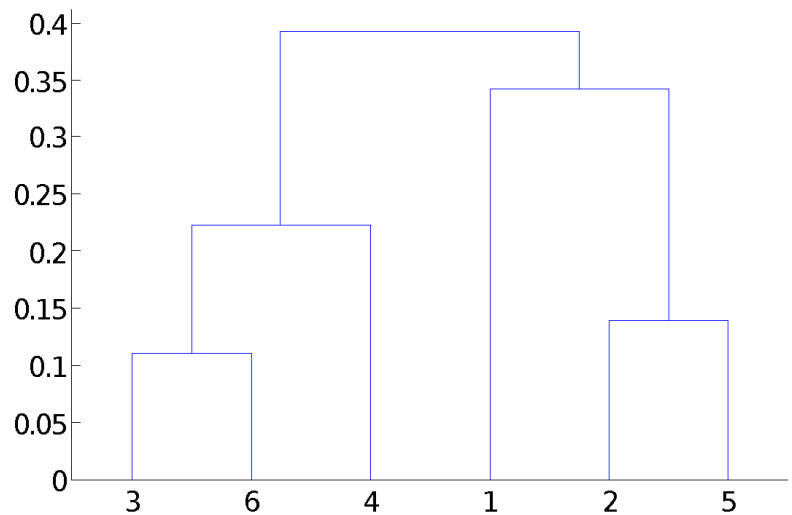
Hierarchical Clustering: MAX



Nested Clusters

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.



Dendrogram

Cluster Similarity: MAX

$$
\begin{aligned}
dist(\{3,6\},\{4\}) &= \max(dist(3,4), dist(6,4)) \\
&= \max(0.15, 0.22) \\
&= 0.22.
\end{aligned}
$$

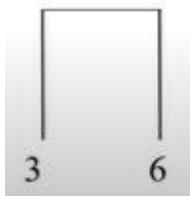
$$dist(\{3,6\},\{2,5\})$$

$$dist(\{3,6\},\{1\})$$

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0



- To update the distance matrix MAX[dist(P3,P6),P1)]
- MAX(dist(P3,P1), (P6,P1))

  = MAX[(0.22,0.23)]

  = 0.23
- To update the distance matrix MAX[dist(P3,P6),P2)]
- MAX(dist(P3,P2), (P6,P2))

  = MAX[(0.15,0.25)]

  = 0.25

	P1	P2	P3	P4	P5
P1	0				
P2	0.23	0			
P3	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0
P6	0.23	0.25	0.11	0.22	0.39

- To update the distance matrix MAX[dist(P3,P6),P4)]
- MAX(dist(P3,P4), (P6,P4))

  = MAX[(0.15,0.22)]

  = 0.22
- To update the distance matrix MAX[dist(P3,P6),P5)]
- MAX(dist(P3,P5), (P6,P5))

  = MAX[(0.28,0.39)]

  = 0.39

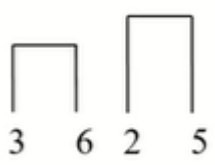
	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

The updated distance matrix for cluster P3,P6

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

From the updated proximity matrix find the minimum distance:
P2,P5

So now update the matrix with respect to these points and update the dendogram too



The distance matrix

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

- To update the distance matrix MAX[dist(P2,P5),P1)]
- MAX[dist(P2,P1), (P5,P1)]

  = MAX[(0.23,0.34)]

  = 0.34
- To update the distance matrix MAX[dist(P2,P5),(P3,P6)]
- MAX[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = MAX[(0.25,0.39)]

  = 0.39

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

- To update the distance matrix MAX[dist(P2,P5),P4)]
- MAX[dist(P2,P4), (P5,P4)]

  = MAX[(0.20,0.29)]

  = 0.29

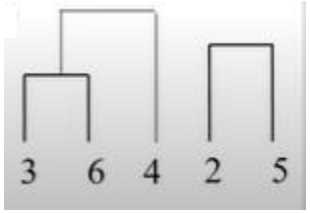
	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

The updated distance matrix for cluster (P2,P5)

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

Find the min value in the matrix: p3,p6 with P4



- To update the distance matrix[((P3,P6),P4) , P1]
- MAX[dist(P3,P6),P1), (P4,P1)]

  = MAX[(0.23,0.37)]

  = 0.37
- To update the distance matrix MAX[dist((P3,P6),P4),(P2,P5)]
- MAX[dist((P3,P6),(P2,P5)), (P4,(P2,P5))]

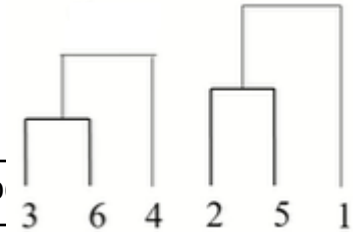
  = MAX[(0.39,0.29)]

  = 0.39

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

The updated distance matrix for cluster P3,P6,P4

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.34	0	
P3,P6,P4	0.37	0.39	0



© Tan,Steinbach, Kumar

- **Cluster** [(P2,P5),P1]
- MAX[dist(P2,P5),(P3,P6,P4), (P1,(P3,P6,P4))]

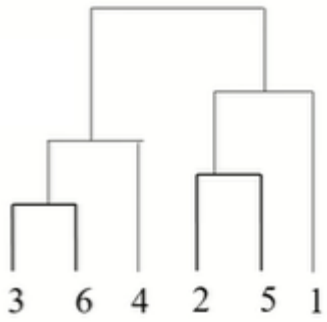
  = MAX[(0.39,0.37)]

  = 0.39

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.54	0	
P3,P6,P4	0.37	0.39	0

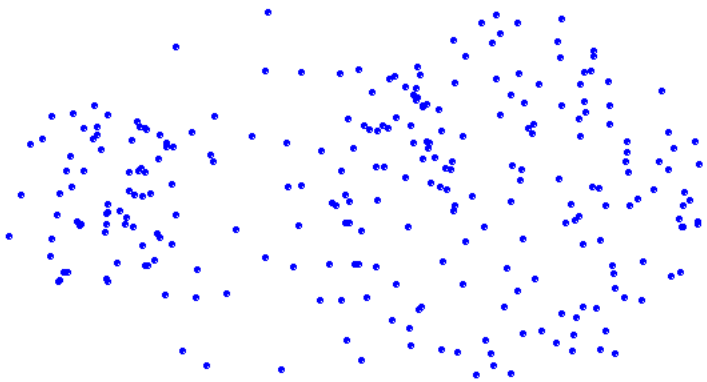
The updated distance matrix for cluster P2,P5,P1

	P2,P5,P1	P3,P6,P4
P2,P5,P1	0	
P3,P6,P4	0.39	0

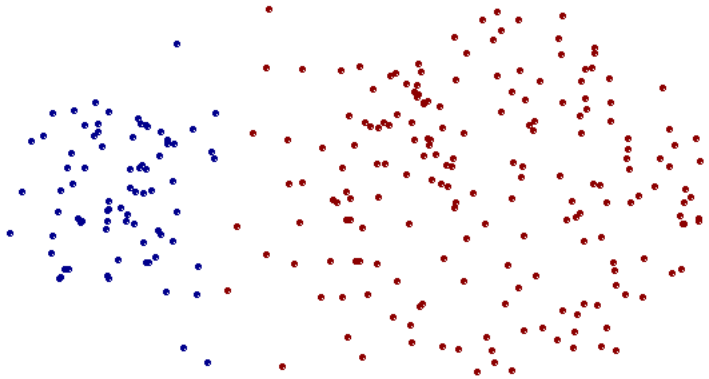


Only one value left so combine it to make one cluster

Strength of MAX



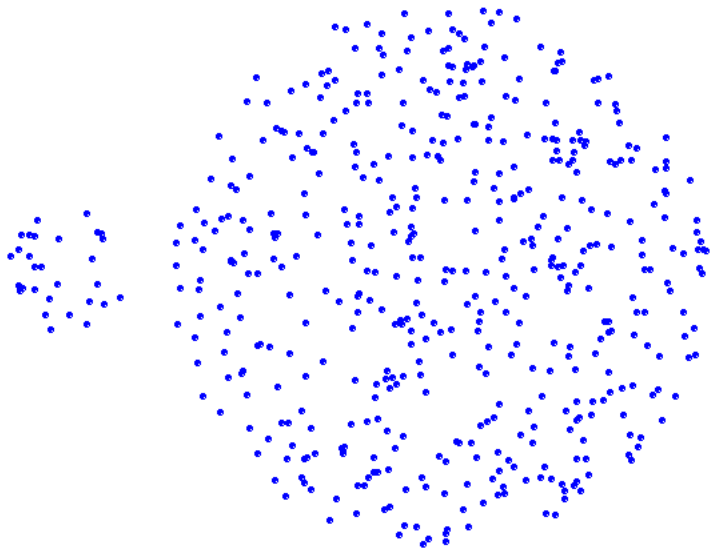
Original Points



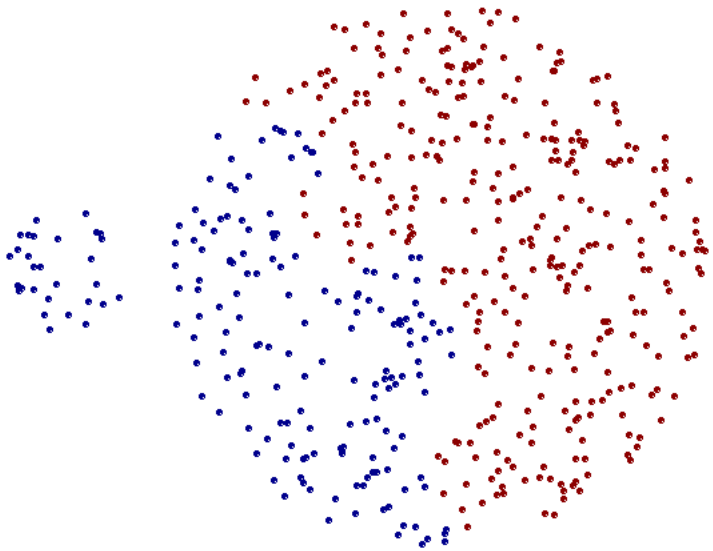
Two Clusters

- **Less susceptible to noise and outliers**

Limitations of MAX



Original Points



Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

Cluster Similarity: Group Average

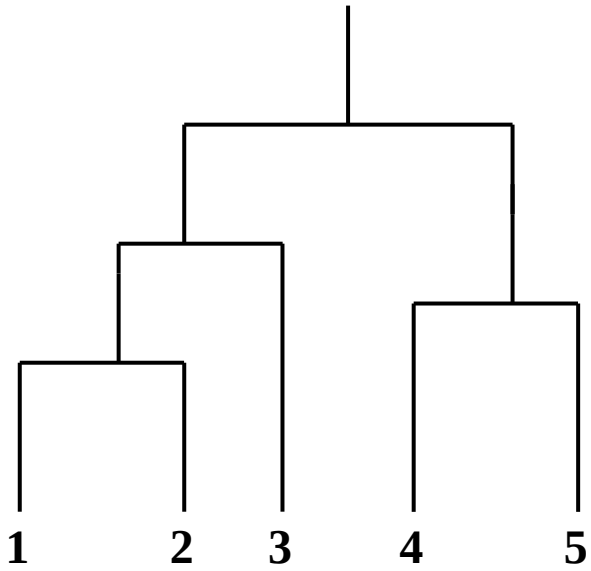
https://www.youtube.com/watch?v=T1ObCUpjq3o

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

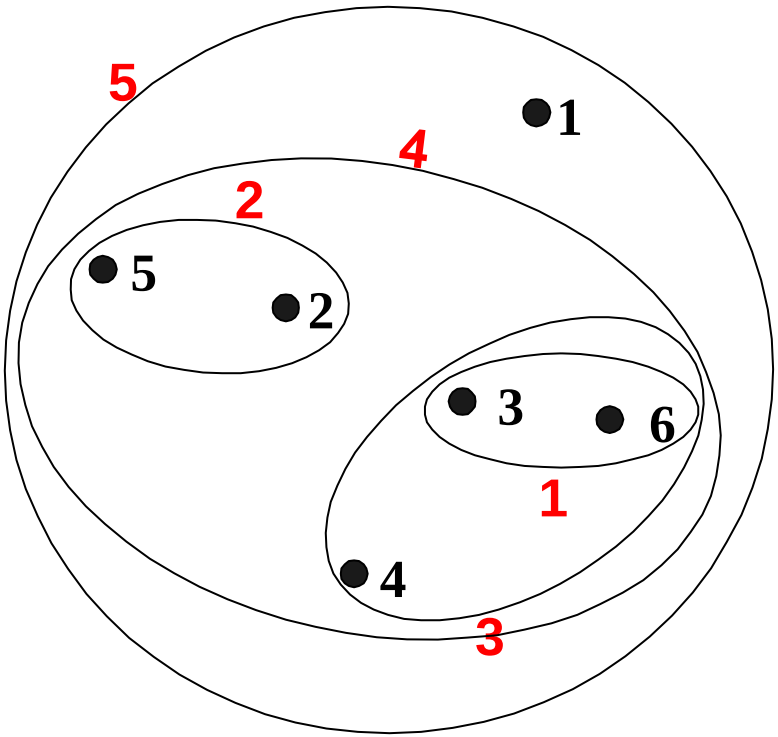
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| * |\text{Cluster}_j|}$$

- Need to use average connectivity for scalability since total proximity favors large clusters

	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



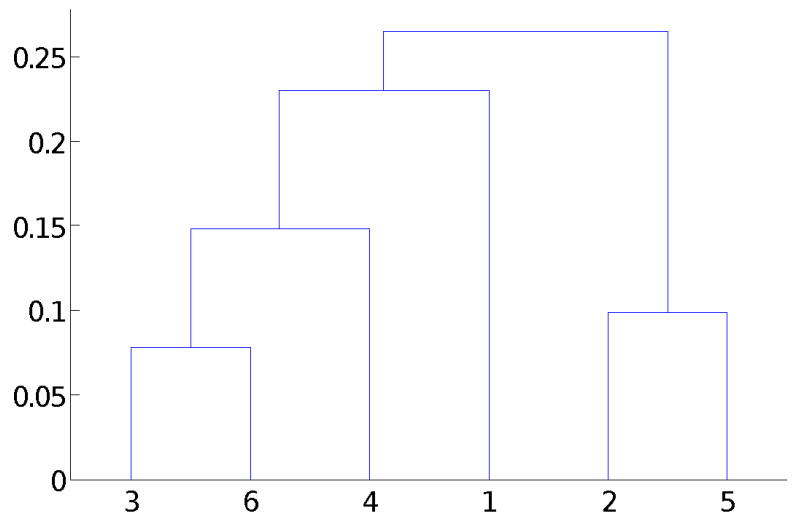
Hierarchical Clustering: Group Average



Nested Clusters

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.



Dendrogram

Cluster Similarity: Group Average

$$dist(\{3,6,4\},\{1\}) = (0.22 + 0.37 + 0.23)/(3 * 1) = 0.28$$

$$dist(\{2,5\},\{1\})$$

$$dist(\{3,6,4\},\{2,5\})$$

The distance matrix is, AVG[dist(P3,P6),P1]

dist((P3,P6),P1) = ½ (dist(P3,P1)+dist(P6,P1)

= ½ (0.22+0.23)

= ½ (0.45)

= 0.23

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Table 8.4. Euclidean distance matrix for 6 points.

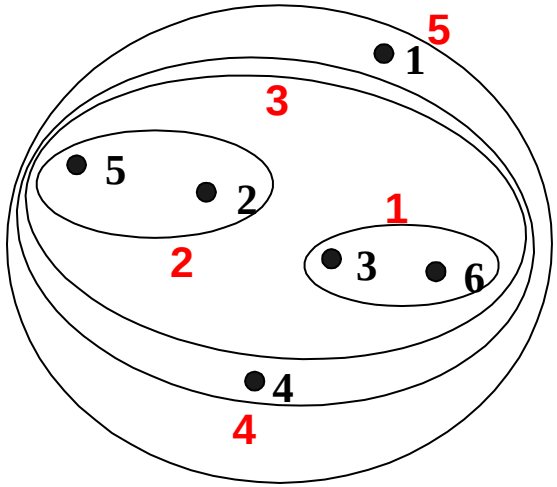
Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise and outliers
- Limitations
  - Biased towards globular clusters

Cluster Similarity: Ward's Method

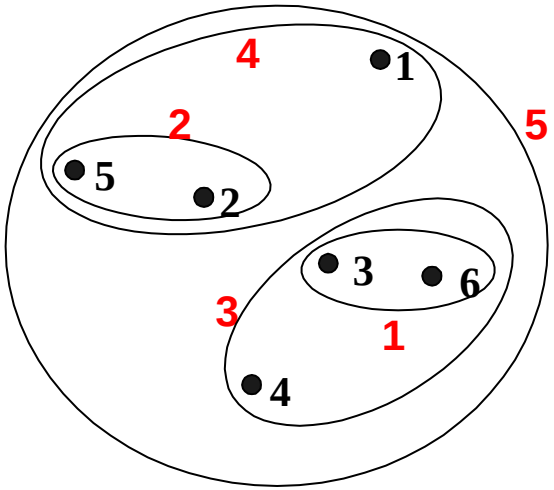
- Similarity of two clusters is based on the increase in squared error when two clusters are merged
  - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
  - Can be used to initialize K-means

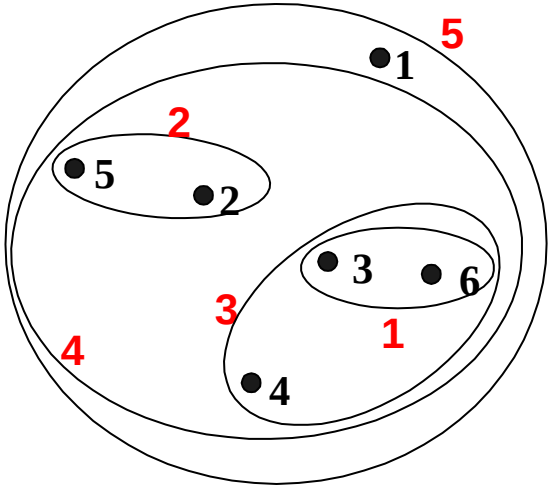
Hierarchical Clustering: Comparison



MIN

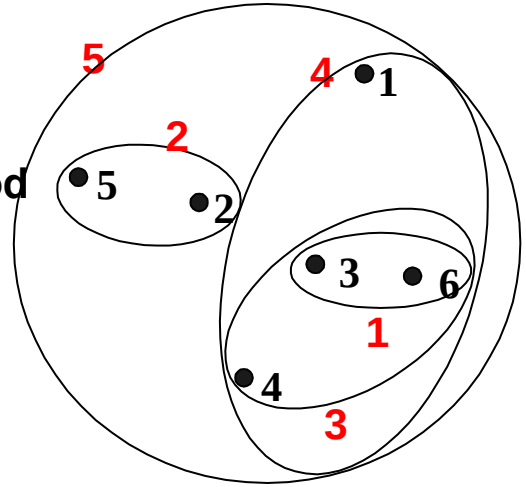
MAX





Group Average

Ward's Method



Hierarchical Clustering: Time and Space requirements

- $O(N^2)$ space since it uses the proximity matrix.
  - N is the number of points.

- $O(N^3)$ time in many cases
  - There are N steps and at each step the size, N^2, proximity matrix must be updated and searched
  - Complexity can be reduced to $O(N^2 \log(N))$ time for some approaches

Hierarchical Clustering: Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No objective function is directly minimized
- Different schemes have problems with one or more of the following:
  - Sensitivity to noise and outliers
  - Difficulty handling different sized clusters and convex shapes
  - Breaking large clusters